Quantitative Empirical Methods Exam

Yale Department of Political Science, August 2025

You have 24 hours to complete the exam. The exam consists of three parts.

Back up your assertions with mathematics where appropriate and show your work. Good answers will provide a direct answer that illustrates an understanding of the question, and calculations or statistical arguments to validate the answer. Where applicable, exceptional answers will include all of these *as well as* proofs that are technically complete, including formally articulating sufficient assumptions and regularity conditions. Questions will not be weighted equally. A holistic score will be assigned to the exam. Therefore, it is important to demonstrate your understanding of the material to the best of your ability.

Part 1 (Theoretical section) consists of short questions that can be answered with pen and paper. You are allowed to consult textbooks and other reference material, but the questions are written so that well-prepared students should be able to answer them without such references.

Part 2 (Essay section) contains a recent, well-regarded empirical article. We will ask you to offer an evaluation of its methodological approach and presentation of results. In particular, we will advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, and possible threats to inference. Your response may be anywhere from 500 to 1,000 words.

Part 3 (Computer assisted section) will involve using statistical software to answer one longer exercise with several associated questions. A complete answer to Part 3 will include code and output, as well as your written answers.

For the whole exam, you are permitted access to any and all written materials, as well as unrestricted use of your own computer with access to the internet. The only restriction is that you may **not** interact with any person, online or otherwise.

Please turn in your answers as an email to colleen.amaro@yale.edu.

1 Theoretical section

1. Provide brief definitions of the following terms:
   (a) p-value
   (b) Sharp null hypothesis
   (c) Stable Unit Treatment Value Assumption

2. Consider an experiment with $2n$ units, with n units receiving the treatment and n units receiving the control, and test the sharp null hypothesis using randomization inference at $\alpha = 0.001$.
   (a) Explain why, in order to have $\alpha = 0.001$, we must have a design that yields at least 1,000 total randomizations (i.e., different ways to assign treatment).
   (b) What is the minimum value of n for a matched pair experiment, so that the smallest p-value does not exceed than 0.001?
   (c) What is the corresponding minimum value of n for a completely randomized experiment?

   Note. If you are having a hard time solving this analytically, you may use simulations to help answer it empirically.

3. Is statistical significance "transitive" in the sense that if A is statistically significantly different from B and B is statistically significantly different from C, A is statistically significantly different from C? If Yes, explain why. If No, give a counterexample.

4. Diana Mutz writes in *Population-Based Survey Experiments*: "It is common practice in many fields of social science to reassure readers that they can trust the internal validity of a given experiment's results by running so-called 'randomization checks,' that is, comparisons between various experimental groups on variables that are not part of the central theoretical framework of the study. Often, these are demographics ...In general, a referee who suggests that an experiment requires a randomization check is probably well meaning, though wrong." On the other hand, Alan Gerber and Donald Green write in *Field Experiments: Design, Analysis, and Interpretation* that "One way to bolster confidence in the integrity of the randomization procedure and the administrative care with which the data were handled is to provide a statistical description of the balance between treatment and control groups on a range of available covariates." What are 2-3 arguments for conducting a randomization check? 2-3 arguments for not conducting a randomization check? Who is right? Why?

5. A rideshare platform enters cities (n = 100) sequentially from 2010-2014. Entry timing depends on: (1) city population thresholds, (2) regulatory approval processes

of varying length, and (3) local taxi industry lobbying intensity. You want to estimate effects on employment in the local transportation sector. You have data on: (1) monthly city-level overall and transportation-specific employment from 2008-2016; (2) platform entry dates and usage volumes by city; (3) city characteristics including a measure on the strictness of taxi regulations, population, income, campaign finance, and election results; and (4) monthly city-level public transit ridership data from 2008-2016. Note that some cities have very strict taxi regulations (e.g., caps on number of taxis allowed to operate in the city) while other cites have very lax regulations (e.g., no caps).

Design your most convincing identification strategy to estimate effects on rideshare platform entry on employment in the local transportation sector. What variation are you exploiting? State your key identifying assumption(s) and explain why they might be violated. How would you test for them given the available data?

6. Scholars have debated the practice of regression adjustment to improve the precision of treatment effect estimates in randomized experiments. It is expected that you will use the internet to research this topic, and your answer should cite relevant readings and references.

   (a) Critically evaluate several arguments for and against the use of regression adjustment with OLS for randomized experiments. (recommended length: about 300 words)

   (b) Under what circumstances is regression adjustment with OLS approximately unbiased?

   (c) Under what circumstances does regression adjustment with OLS improve precision?

7. You are asked to review a paper that is examining the impact of algorithmically generated risk recommendations to judges' sentencing decisions on whether or not to impose cash bail for individuals in pre-sentencing trials. The algorithm is deterministic, and takes in a risk score that ranges from 1 to 6, where 1 is considered low-risk and 6 is high-risk. A risk score above a 3 will result in the algorithm recommending cash bail, whereas a risk score below a 3 will result in the algorithm recommending release. The authors want to know whether a "cash bail" recommendation from the algorithm will change the sentencing decision of the judge. The outcome of interest here is the sentencing decision of the judge, where $Y = 1$ for sentencing with cash bail and 0 implies release.

   (a) The authors begin by using a sharp regression discontinuity design. They report a treatment effect of 0.3. What is the estimand of interest in an RDD? What target population does the RDD estimate for?

(b) Explain in words how you would run this RDD. What is the running variable?

(c) What assumptions must be true for this to be a valid RDD?

(d) Explain the different researcher degrees-of-freedom that can impact the downstream analysis.

(e) The authors do not provide any robustness checks. What are some robustness checks the authors should provide to help reason about the validity of their underlying RDD assumptions? Name 2 examples of robustness checks, and how they would help improve the credibility of their RDD estimate.

(f) In the next part of the paper, the authors run a regression, where they regress the judges' sentencing decisions with the algorithmically generated risk recommendations, and a set of pre-treatment covariates, such as the demographic information about the cases in the study and judges fixed effects. They find a statistically insignificant treatment effect, with point estimate of 0.2. They conclude that even though it is not significant, the point estimate is positive, and validates the RDD result. Is this a valid claim? Explain why or why not.

2 Essay section

Read the following article and offer a critical evaluation of its methodological approach and presentation of results. Note: "critical" does not imply that you must only criticize – it is recommended that you give credit to the authors if and when their arguments are convincing and/or novel with respect to standard practice.

Your response should be between 500 to **1000** words. We advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, possible threats to inference, statistical power, and generalizability. Justify each of your claims and, where applicable, suggest ways in which this line of research might be improved. We do not expect you to have special expertise in the topic area, but we do expect you to bring to bear your general analytical skills as a political scientist.

Article:

> Myers, Andrew C.W. "Press Coverage and Accountability in State Legislatures." *American Political Science Review* (Forthcoming). `https://doi.org/10.1017/S000305542500022X`.

3 Computer assisted section

Construct a function `generate_data` that takes, as an input, a sample size n, and does the following:

- Generates a set of 10 covariates: $X_1, \ldots, X_{10}$, where the first five are drawn from a standard Normal distribution, and the last 5 are Bernoulli random variables.
- Generate the probability of treatment as a logistic function of the linear combination of covariates X_1, X_2, and X_9
- Using the generated treatment probabilities, generate a vector of treatments Z, where $Z = 1$ implies treatment, and $Z = 0$ implies control (Hint: use `rbinom`)
- Construct an outcome model, where Y is a linear function of all 10 covariates, with coefficients $\beta = (1, 2, -0.3, -0.4, 1, 0.9, 2, -0.1, 2, 1)$:

  $$Y = \tau \cdot Z + \sum_{p=1}^{10} \beta_p X_p + \epsilon,$$

  where $\tau = 2$, and $\epsilon \sim N(0, 1)$.
- Outputs a data frame, with all 10 covariates, the observed Y, $Y(1)$, $Y(0)$, and Z.

For sample sizes $\{100, 1000, 5000, 10000\}$:

- Use `generate_data` to simulate a dataset
- Estimate the following:
  - A simple difference-in-means between the treatment/control groups
  - An IPW estimator using:
    - All 10 covariates
    - Only covariates X_1 and X_2
    - Covariates X_1, X_2, X_9

    Propensity scores should be estimated using logistic regression.
  - An OLS estimator, where in the regression, you adjust for:
    - All 10 covariates
    - Only covariates X_1 and X_2

    * Covariates X_1, X_2, X_9

- For each sample size, simulate 1,000 iterations, and then summarize the results:
  - Generate a table that provides the mean and standard deviation of each of the estimators for each sample size
  - Create a figure, containing boxplots of the different estimates across the different simulation settings.
- What do you takeaway from the simulation results? We have simulated the performance of IPW and OLS estimators using different estimation sets (i.e., different sets of covariates). Explain how the results differ by the estimation sets used.